

delicious but also has all the right nutritional values. You leave for your office in your self-driving car and reach on time. Traffic isn't a problem anymore in a world where all cars communicate with each other. They calculate the optimum route and speed to reach their destinations. Your workstation boots up as soon as you enter and you sit down to work on your current project – a robot that helps you make more robots. Seems like a stretch?

Back to 2017, we are slowly but steadily moving toward a completely automated future. Take a look around you, technology giants are working hard to make self-driving cars a reality, and they are really, really close. New innovations and breakthrough technologies are coming out almost every day. If you had told someone 30 years ago that they could see and talk to anyone across the globe in real time, they probably wouldn't have believed you, in fact maybe even calling you just another nutjob. Nonetheless, the future is here now and the magical thing that is changing our lives currently are 'robots'.

What defines a robot?

According to Wikipedia, a machine is any mechanical device that “changes the direction or magnitude of a force”. Machines can be divided into six simple categories:

1. Lever
2. Wheel and axle
3. Pulley
4. Inclined plane
5. Wedge
6. Screw



A machine can be any of these six or a combination of two or more that performs a certain task. A robot is, well, much more complicated. It can be a combination of machines such that:

- It can be programmed
- Interacts with the environment and takes in 'perceptions'
- Processes data and works autonomously to some extent
- Capable of making changes to its physical environment

The main difference between a machine and a robot is that robots are 'smarter' in the sense that a machine is capable of only performing a set of